
Cleared for Takeoff? A Visual Analysis of U.S. Domestic Flight Delay Patterns (2019–2023)

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Abstract

This paper presents a visual analysis of U.S. domestic flight delay patterns using five years of on-time performance data from the Bureau of Transportation Statistics (2019–2023). The analysis is structured as a four-chapter data story examining delay patterns through the lenses of time, geography, delay cause, and airline performance. Twelve visualizations were produced using a combination of `matplotlib`, `seaborn`, `Altair`, and `Plotly`. Key findings challenge common assumptions: airport size does not predict delay rate, weather ranks last among delay causes despite public perception, and low-cost carriers dominate the worst-performers list regardless of their independence from major airline scheduling. The analysis concludes with actionable takeaways relevant to travelers, airline operations teams, and airport planners.

1. Introduction

Flight delays cost the U.S. economy an estimated \$33 billion per year in direct and indirect costs (Federal Aviation Administration, 2020), touching nearly every American traveler. Despite their frequency, the patterns behind delays are routinely misunderstood — most passengers attribute delays to weather or air traffic control when data from the Bureau of Transportation Statistics (BTS) reveals a more nuanced picture dominated by factors within airline control.

This work addresses the following four central questions through exploratory visual analysis:

Q1. When do delays peak? (temporal patterns)

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Q2. Where are delays most severe? (geographic patterns)

Q3. Why do flights get delayed? (delay cause breakdown)

Q4. Who is most affected? (airline and route analysis)

Rather than building a predictive model, this analysis takes a step back to let the data speak through carefully designed visualizations — prioritizing clarity, interpretability, and insight over technical complexity.

The remainder of this paper is organized as follows. Section 2 describes the dataset and preparation steps. Section 3 details the methods and tools used. Section 4 discusses visualization design and rationale. Section 5 presents key findings organized by chapter. Section 6 discusses limitations and future work.

2. Dataset Description

2.1. Source and Scale

The dataset used is the Flight Delay and Cancellation Dataset 2019–2023, sourced from the U.S. Department of Transportation’s Bureau of Transportation Statistics and distributed via Kaggle (Zel, 2023). The BTS is the authoritative federal source for commercial aviation performance data.

Table 1 summarizes the dataset attributes.

Table 1. Dataset Summary

Attribute	Detail
Time Span	January 2019 – August 2023
Raw Size	~3 million rows, 32 columns
Format	CSV
Access	Kaggle API via <code>kagglehub</code>
Delay Rate	~18.2% of flights

Key columns used in the analysis include: `FL_DATE`, `AIRLINE_CODE`, `ORIGIN`, `DEST`, `CRS_DEP_TIME`, `DEP_DELAY`, `DISTANCE`, and the five delay cause columns (`DELAY_DUE_CARRIER`, `DELAY_DUE_WEATHER`, `DELAY_DUE_NAS`, `DELAY_DUE_SECURITY`, `DELAY_DUE_LATE_AIRCRAFT`).

2.2. Data Preparation

The following preparation steps were applied prior to analysis:

(1) Canceled flights removed. Rows where `DEP_DELAY` is null were dropped, as cancellations represent a qualitatively different outcome from delays.

(2) Binary delay flag engineered. `IS_DELAYED` was constructed from `DEP_DELAY` using the FAA/DOT standard of ≥ 15 minutes as the threshold for a reportable delay:

$$\text{IS_DELAYED} = [\text{DEP_DELAY} \geq 15] \quad (1)$$

(3) Temporal features extracted. `MONTH`, `DAY_OF_WEEK`, `YEAR`, and `HOUR_OF_DAY` were derived from `FL_DATE` and `CRS_DEP_TIME`.

(4) Weekend flag engineered. `IS_WEEKEND` was derived from `DAY_OF_WEEK` to simplify weekly pattern analysis.

(5) Column reference standardized. A single control-panel cell defines all column name variables, making the notebook resilient to minor schema differences across dataset versions.

3. Methods and Tools

3.1. Environment

The analysis was conducted in a Google Colab notebook environment using Python 3. The dataset was loaded programmatically via the `kagglehub` library, eliminating the need to attach a CSV file and making the notebook fully shareable — any user with a free Kaggle account can authenticate and run the notebook end to end.

3.2. Libraries

Table 2 summarizes the libraries used and their specific roles in the analysis.

Table 2. Libraries Used

Library	Role
<code>pandas</code>	Data loading, cleaning, aggregation
<code>numpy</code>	Numerical operations, CI computation
<code>matplotlib</code>	Static bar charts, stacked charts
<code>seaborn</code>	Heatmaps, box plots, grouped bars
<code>altair</code>	Interactive line, area, scatter charts
<code>plotly</code>	Interactive geospatial bubble map
<code>kagglehub</code>	Dataset acquisition from Kaggle

3.3. Analytical Approach

All analysis is descriptive and exploratory. Aggregations were computed using `pandas groupby` operations, and

chart types were selected based on the structure of the data being visualized. For the weather delay seasonality analysis, the analysis was restricted to flights with a recorded weather delay greater than zero minutes:

$$\mu_{\text{weather}} = \frac{1}{|S|} \sum_{i \in S} d_i, \quad S = \{i : \text{DELAY_DUE_WEATHER}_i > 0\} \quad (2)$$

where d_i is the weather delay in minutes for flight i . This prevents the per-flight average from being diluted by flights with no weather delay component, yielding a more meaningful measure of delay severity conditional on weather being the cause.

Airport geolocation data was sourced at runtime from a public airport codes dataset (Datasets Contributors, 2024) and merged on IATA code to support the bubble map visualization.

4. Visualization Design and Rationale

Each visualization was selected to answer a specific question rather than simply display a metric. Table 3 summarizes all twelve charts produced.

Table 3. Visualization Inventory

Chart	Type	Library
Monthly delay heatmap	Heatmap	seaborn
Delay by hour of day	Line + CI band	Altair
Delay by day of week	Bar chart	seaborn
Airport bubble map	Geo scatter	Plotly
Top 20 delayed routes	H. bar chart	matplotlib
Houston spotlight	Annotated bar	matplotlib
Cause by year	100% stacked	matplotlib
Cause by airline	Grouped bar	seaborn
Weather delay seasonality	Layered area	Altair
Airline report card	Ranked H. bar	matplotlib
Delay vs. distance	Scatter + trend	Altair
Delay distribution	Box plot	seaborn

4.1. Temporal Charts

The monthly heatmap (Section 1.1) encodes two dimensions simultaneously — month and year — allowing identification of both seasonal cycles and year-over-year structural changes in a single view. Percentage annotations prevent misreading due to color alone.

The hourly line chart (Section 1.2) uses Altair for its interactive tooltip capability. The confidence band was set at ± 1 standard deviation rather than the standard 95% interval (± 1.96 SD) to maintain visual visibility at high sample sizes, where the narrower interval renders as a hairline.

4.2. Geographic Charts

Plotly's interactive `Scattergeo` was used for the airport bubble map (Section 2.1) to enable hover tooltips with airport code, delay rate, and flight volume. Bubble size encodes flight volume and color encodes delay rate, allowing both dimensions to be read simultaneously. A horizontal bar chart was selected for the route ranking (Section 2.2) because route labels are long and would require rotation on a vertical axis.

4.3. Cause and Airline Charts

A 100% stacked bar chart (Section 3.1) normalizes for total flight volume across years, enabling direct comparison of delay cause *mix* rather than raw totals — critical given the large volume reduction during COVID-19. A box plot was selected for the delay distribution (Section 4.3) over a bar chart specifically because it reveals median, spread, and outliers simultaneously, which a mean-based chart would obscure.

5. Key Findings

5.1. Chapter 1: Temporal Patterns

Two seasonal peaks dominate the five-year record: summer (June–August) and the winter holiday corridor (November–January). The most striking feature is 2020 — delay rates collapse to near zero beginning in March, the month COVID-19 was declared a global pandemic, visible as early as January 2020. *See Figure 1.*

The hourly analysis reveals a strong positive trend from 5:00 AM (7.3% delay rate) to 9:00 PM (28.3% delay rate), consistent with a compounding domino effect in which early morning delays propagate downstream through aircraft rotations and crew schedules as shown in *Figure 2.*

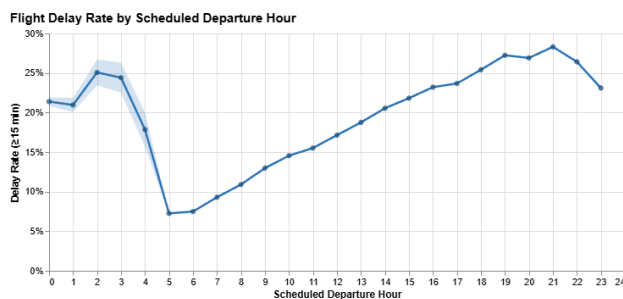


Figure 2. Flight Delay Rate by Scheduled Departure Hour.

At the day-of-week level, only a 3.7 percentage point spread separates Tuesday (best, 17.1%) from Friday (worst, 20.8%), confirming that time of day and time of year are far stronger predictors than day of week. *Figure 3.*

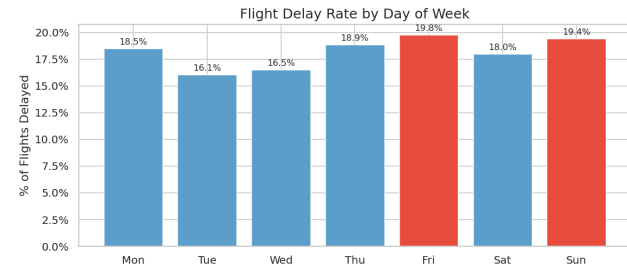


Figure 3. Flight Delay Rate by Day of Week.

5.2. Chapter 2: Geographic Patterns

Across 237 mapped U.S. airports, airport size is not a reliable predictor of delay rate. Large high-traffic hubs and smaller regional airports frequently perform at similar delay rates, suggesting that operational efficiency and geographic exposure matter more than volume alone. *See Figure 4.*

At the route level, Chicago Midway (MDW) accounts for 7 of the 20 most-delayed routes in the country — nearly double the contribution of any other single airport. For the Houston market specifically, William P. Hobby Airport (HOU) carries a 24.3% departure delay rate, placing it among the worst-performing airports nationally, while George Bush Intercontinental Airport (IAH) sits at the national average of 18.2%.

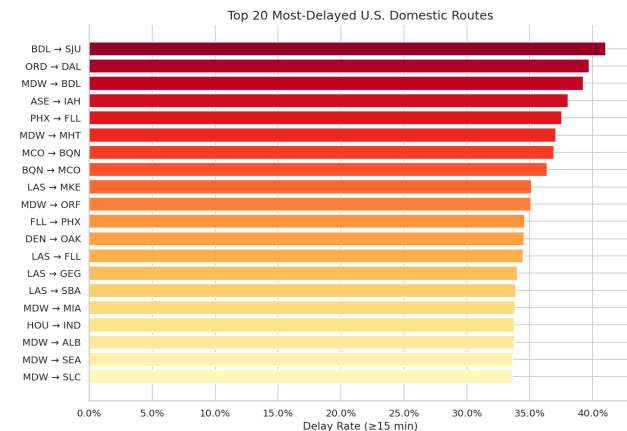


Figure 5. Top 20 Most Delayed Routes.

5.3. Chapter 3: Delay Cause Breakdown

Weather is not the primary culprit. Across all five years, Carrier and Late Aircraft delays dominate the composition chart, while Weather ranks as the least common cause, ahead of only Security. Late Aircraft delays are particularly significant because they are the most preventable — when an incoming aircraft arrives late, every subsequent flight assigned to that plane absorbs the delay in a cascading chain reaction.

Cleared for Takeoff? Visual Analysis of U.S. Flight Delay Patterns

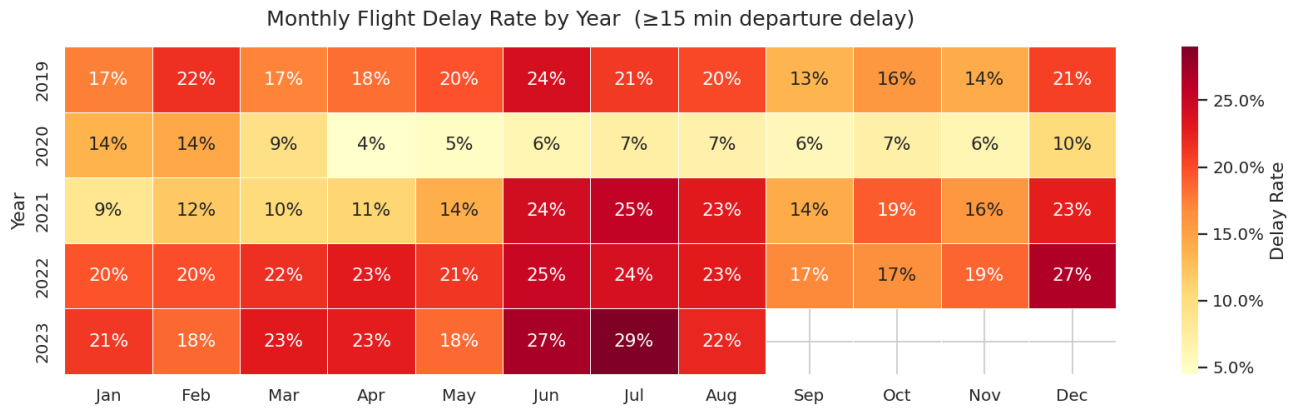


Figure 1. Heatmap: Monthly Flight Delay Rate by Year.

Departure Delay Rate by Origin Airport (2019–2023)
Bubble size = flight volume | Color = delay rate (yellow = low, red = high)

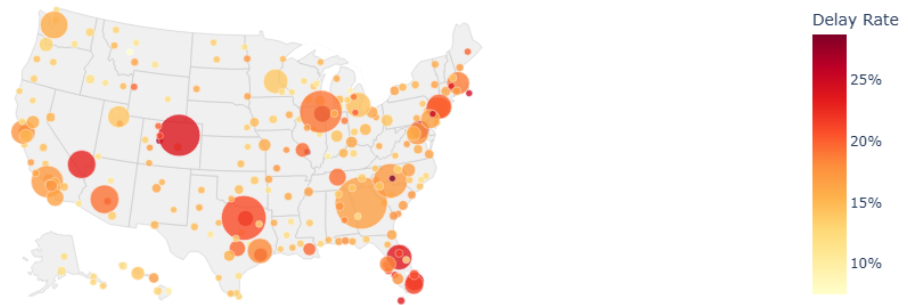


Figure 4. Bubble Map: Departure Delay Map by Origin Airport.

The most striking individual finding is SkyWest Airlines (OO), whose Carrier delay minutes are approximately double those of any other carrier. This likely reflects scheduling pressure inherited from the mainline carriers (Delta, United, American) that SkyWest operates for, rather than SkyWest's own management failures. *See Figure 6.*

When weather does cause a delay, the severity is consistent across seasons: average weather delay length ranges between 64 and 73 minutes with no dramatic seasonal spike. Summer thunderstorms produce delays comparable in severity to winter storms.

5.4. Chapter 4: Airline and Route Analysis

The five worst-performing carriers by delay rate — Jet-Blue, Frontier, Allegiant, Southwest, and Spirit — are all fully independent low-cost or ultra-low-cost carriers (LCCs/ULCCs). Their elevated delay rates reflect the ag-

gressive turnaround times and high aircraft utilization that characterize the LCC business model, making them disproportionately vulnerable to the domino effect identified in Chapter 1.

Flight distance is a poor predictor of delay risk. The scatter plot reveals only a very weak positive relationship between route distance and delay rate. The distribution analysis reveals that what separates good carriers from poor ones is not only how often they delay, but how long and how consistently: Hawaiian Airlines (HA) exhibits the tightest, lowest delay distribution, while ExpressJet (EV) and Mesa (YV) show the widest spread, indicating high unpredictability in addition to elevated severity. *See Figure 7.*

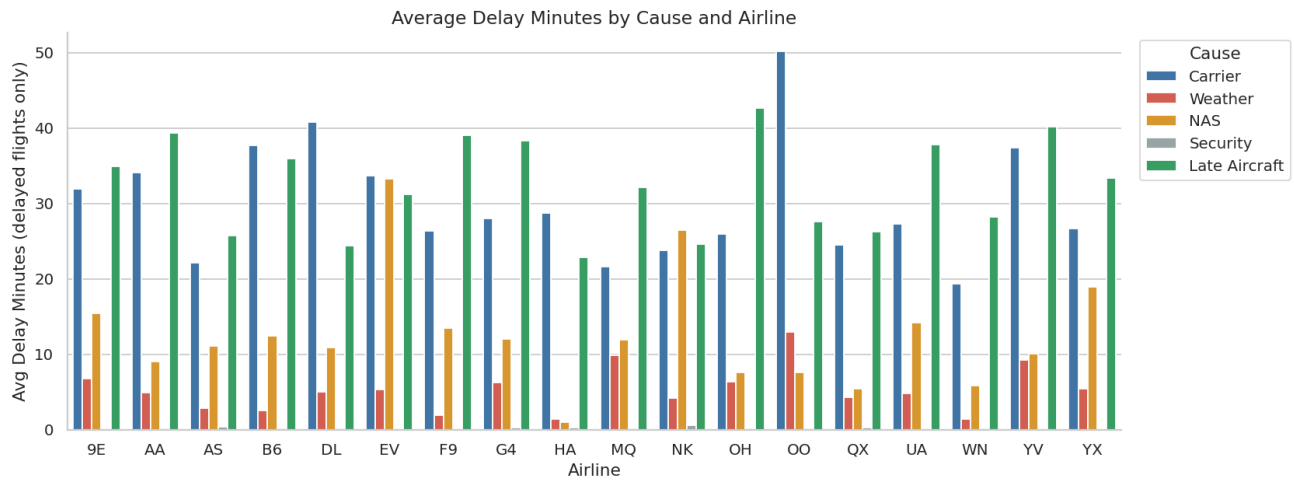


Figure 6. Grouped Bar: Avg Delay Minutes by Cause and Airline.

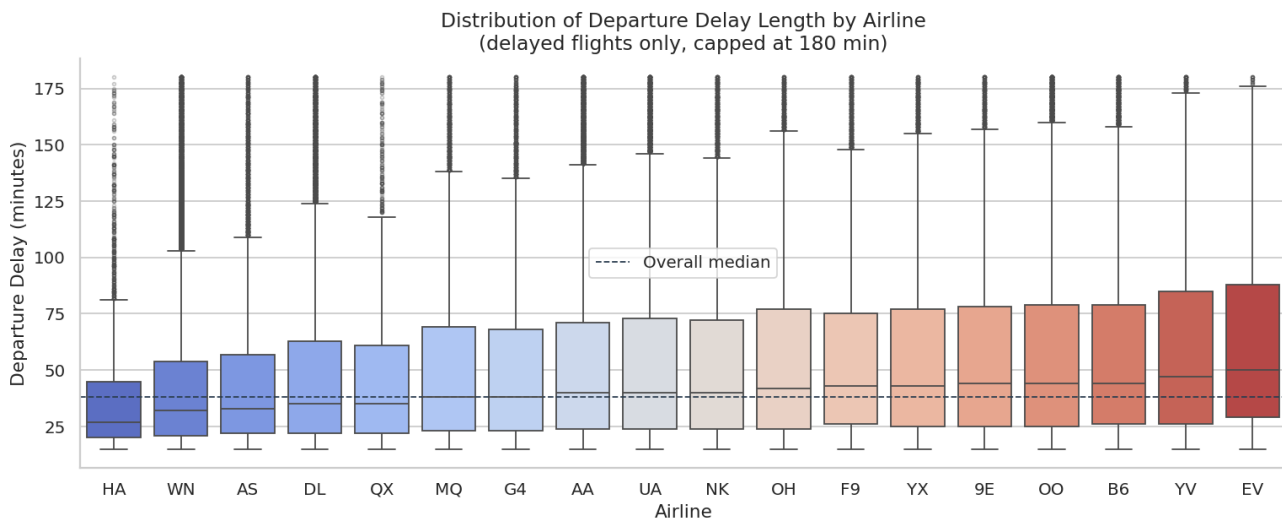


Figure 7. Boxplot: Distribution of Departure Delay Length by Airline.

6. Limitations and Future Improvements

6.1. Limitations

Absence of real-time weather data. The dataset includes `DELAY_DUE.WEATHER` as a post-hoc attribution rather than actual meteorological conditions. Actual weather variables (wind speed, precipitation, visibility) are not available, limiting the depth of causal analysis in Chapter 3.

COVID-19 structural break. The 2020 data represents a fundamentally abnormal operating environment — flight volumes fell by over 60% and delay patterns reflect extraordinary circumstances. Including 2020 alongside surrounding years introduces a structural break that can obscure underlying trends in charts not specifically designed to highlight it.

Geospatial data dependency. The bubble map depends on a runtime merge with an external airport coordinates dataset (Datasets Contributors, 2024). Changes to this external source’s format or availability would break the visualization.

Carrier code readability. The dataset stores airline identifiers as two-letter IATA codes. Human-readable airline names are provided in companion reference tables rather than in the charts themselves, requiring cross-referencing.

6.2. Future Improvements

Integrate meteorological data. Joining historical weather observations by airport and date would allow Chapter 3 to distinguish genuinely weather-driven delays from those airlines attribute to weather for operational reasons.

Separate pre- and post-COVID analysis. Splitting into two periods (2019 and 2021–2023) would yield cleaner trend analysis, treating 2020 as a separate case study in systemic disruption.

On-time recovery analysis. A flight that departs late but arrives on time due to schedule padding or favorable conditions represents a meaningfully different outcome. Incorporating arrival delay alongside departure delay would provide a more complete picture of actual passenger impact.

Interactive dashboard. Converting the notebook into a Streamlit or Dash web application would allow end users to filter dynamically by airline, airport, or date range, transforming the analysis into a practical operational tool.

Expand the Houston spotlight. The Houston-specific section could be deepened to include delay patterns by time of day, cause breakdown by airport, and comparison against other major Texas airports (DFW, DAL, AUS, SAT) for regional context.

7. Conclusion

This analysis demonstrates that the conventional wisdom surrounding flight delays — that weather and air traffic control are the primary culprits — is not supported by five years of BTS performance data. The dominant causes of delay are Carrier and Late Aircraft, both attributable to airline operations rather than external forces. Temporal patterns reveal a strong domino effect that compounds throughout the day, and geographic analysis shows that delay risk clusters around specific origin airports rather than tracking with airport size. Together, these findings suggest that the most impactful interventions for reducing U.S. domestic flight delays lie within airline operational control — particularly Late Aircraft turnaround efficiency — rather than in infrastructure investment or weather mitigation.

References

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